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Injection-moldable aerator mixing rod and method of manufacturing thereof

Abstract

An aerator mixing rod includes a body having a first and second opposite ends and an outer surface extending between the ends. The body defines a longitudinal axis extending through the ends. The aerator mixing rod also includes teeth extending radially outward from the outer surface, with a first row of teeth and a second row of teeth spaced from the first row along the longitudinal axis. A first passageway is formed between adjacent teeth of the first row, and a second passageway is formed between adjacent teeth of the second row. The first passageway is at least partially misaligned with the second passageway in a direction parallel to the longitudinal axis such that the first passageway and the second passageway form a tortuous path for fluid flowing along the outer surface of the body. The aerator mixing rod is formed from an injection-moldable material by an injection molding process.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application No. 63/025,283, filed May 15, 2020, the entire content of which is incorporated herein by reference.

FIELD OF THE DISCLOSURE
(1) The present disclosure relates to an aerator mixing rod, and more particularly to an injection-molded aerator mixing rod.
BACKGROUND OF THE DISCLOSURE
(2) An aerator mixing rod is a static mixing rod that mixes and aerates a liquid product as the product flows over or through the mixing rod. Such aerator mixing rods may be incorporated into food product dispensers, such as whipped product (e.g., whipped cream or other aerated emulsions) dispensers. Typical whipped product dispensers may include a product reservoir containing a liquid product to be whipped, a whipping assembly (including the aerator mixing rod), and a drive mechanism configured to move product from the product reservoir through the whipping assembly to form a whipped product. The whipped product is then dispensed through a nozzle for use.
(3) To emulsify the liquid product and form whipped cream from the whipping assembly, the drive mechanism forces the liquid product through passageways formed between teeth on an outer surface of the aerator mixing rod. When the liquid product moves through the passageways on the aerator mixing rod, air/pressurized gas mixes with the liquid product to form a foam-like airy substance.
(4) Due to their complex geometries, known aerator mixing rods are manufactured using either CNC machining, an Electric Discharge Machine (EDM), or investment casting. However, these conventional methods are proven to be costly and time intensive when producing aerator mixing rods on an industrial scale. Additionally, these methods may yield aerator mixing rods with dimensional inconsistencies or overall unsatisfactory surface finishes, resulting in waste and increasing overall costs.
SUMMARY OF THE DISCLOSURE
(5) The present disclosure provides, in a first aspect, an aerator mixing rod including a body having a first end, a second end opposite the first end, and an outer surface extending between the first and second ends. The body defines a longitudinal axis extending through the first and second ends. The aerator mixing rod also includes a plurality of teeth extending radially outward from the outer surface of the body, with a first row of teeth and a second row of teeth spaced from the first row of teeth along the longitudinal axis. A first passageway is formed between adjacent teeth of the first row of teeth, and a second passageway is formed between adjacent teeth of the second row of teeth. The first passageway is at least partially misaligned with the second passageway in a direction parallel to the longitudinal axis such that the first passageway and the second passageway are configured to at least partially form a tortuous path for fluid flowing along the outer surface of the body. The aerator mixing rod is formed from an injection-moldable material by an injection molding process.
(6) In some aspects, each tooth of the plurality of teeth includes a first side and a second side

generally opposite the first side, and the first and second sides converge in a radial outward direction of the aerator mixing rod.
(7) In some aspects, the first and second sides are axially-facing sides.
(8) In some aspects, the first and second sides are circumferentially facing sides.
(9) In some aspects, each tooth of the plurality of teeth includes an inner width and an outer width. The inner width is defined adjacent the outer surface of the body, and the inner width is greater than the outer width.
(10) In some aspects, the inner width is at least 5% greater than the outer width.
(11) In some aspects, each tooth of the plurality of teeth includes an outermost point. The outermost points of the plurality of teeth collectively define a first diameter, the body defines a second diameter, and the second diameter is between 10% and 95% of the first diameter.
(12) In some aspects, the injection-moldable material is a plastic material.
(13) In some aspects, the plastic material includes an acetal homopolymer.
(14) In some aspects, the first row of teeth and the second row of teeth each include 8 teeth.
(15) In some aspects, the aerator mixing rod includes a core and an outer shell, the outer shell is molded over the core, and the outer shell includes the plurality of teeth.
(16) The present disclosure provides, in a second aspect, a food product dispenser including a drive unit and a dispensing unit coupled to the drive unit. The dispensing unit includes a product reservoir configured to store a food product, a dispensing nozzle, and a product transfer assembly including an aerator mixing rod. The product transfer assembly is configured to be driven by the drive unit to convey the food product from the product reservoir, along the aerator mixing rod, and to the dispensing nozzle, and the aerator mixing rod is formed from an injection-moldable material by an injection molding process.
(17) In some aspects, the aerator mixing rod includes a thermal conductivity between 0.1 and 0.5 Watts/Meter-Kelvin.
(18) In some aspects, the aerator mixing rod includes a first portion and a second portion arranged side-by-side within the product transfer assembly.
(19) In some aspects, the aerator mixing rod includes a first portion and a second portion arranged concentrically within the product transfer assembly.
(20) In some aspects, the injection-moldable material is a plastic material.
(21) In some aspects, the injection-moldable material includes an acetal homopolymer.
(22) In a third aspect, the present disclosure provides a method of manufacturing an aerator mixing rod using injection molding. The method includes arranging a plurality of mold parts into a mold assembly structure, the mold assembly structure defining a mold cavity shaped to form the aerator mixing rod such that the aerator mixing rod includes a body having a first end, a second end opposite the first end, and an outer surface extending between the first and second ends. The body defines a longitudinal axis extending through the first and second ends. A plurality of teeth extends radially outward from the outer surface of the body, the plurality of teeth including a first row of teeth and a second row of teeth spaced from the first row of teeth along the longitudinal axis. A first passageway is formed between adjacent teeth of the first row of teeth, and a second passageway is formed between adjacent teeth of the second row of teeth. The first passageway is at least partially misaligned with the second passageway in a direction parallel to the longitudinal axis such that the first passageway and the second passageway are configured to at least partially form a tortuous path for fluid flowing along the outer surface of the body. The method further includes heating an injection moldable material, injecting the injection moldable material into the mold assembly structure to form the aerator mixing rod, and, after a cooling period, removing the aerator mixing rod from the mold assembly structure by separating the plurality of mold parts from the aerator mixing rod.
(23) In some aspects, arranging the plurality of mold parts into the mold assembly structure includes selecting between a plurality of different mold release configurations in order to prevent

an undercut.
(24) In some aspects, at least one of the mold release configurations is a symmetrical configuration including a plurality of part lines intersecting at a center of the mold cavity, and wherein separating the plurality of mold parts includes separating the mold parts along the part lines.
(25) In some aspects, least one of the mold release configurations is an asymmetrical configuration including four part lines, and wherein separating the plurality of mold parts includes separating the mold parts along the part lines.
(26) Other features and aspects of the disclosure will become apparent by consideration of the following detailed description and accompanying drawings. Any feature(s) described herein in relation to one aspect or embodiment may be combined with any other feature(s) described herein in relation to any other aspect or embodiment as appropriate and applicable.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view of an aerator mixing rod according to an embodiment of the disclosure.
(2) FIG. **2** is a longitudinal cross-sectional view of the aerator mixing rod of FIG. **1**.
(3) FIG. **3**A is an enlarged side view illustrating a portion of the aerator mixing rod of FIG. **1**.
(4) FIG. **3**B is an enlarged perspective view illustrating a portion of the aerator mixing rod of FIG. **1**.
(5) FIG. **4** is a transverse cross-sectional view of the aerator mixing rod of FIG. **1**.
(6) FIG. **5**A illustrates a molding assembly having a symmetrical release configuration, which may be used in molding an aerator mixing rod such as the aerator mixing rod of FIG. **1**.
(7) FIG. **5**B illustrates a molding assembly having an asymmetrical mold release configuration, which may be used in molding an aerator mixing rod such as the aerator mixing rod of FIG. **1**.
(8) FIG. **6**A is an end view of an aerator mixing rod illustrating an asymmetrical mold release configuration that results in an undercut.
(9) FIG. **6**B is an end view of the aerator mixing rod of FIG. **6**A illustrating a symmetrical mold release configuration that eliminates the undercut of FIG. **6**A.
(10) FIG. **7**A is an end view of an aerator mixing rod having a first tooth geometry.
(11) FIG. **7**B is an end view of an aerator mixing rod having a second tooth geometry.
(12) FIG. **8** is a perspective view of an aerator mixing rod according to another embodiment.
(13) FIG. **9**A is a perspective view illustrating a portion of an aerator mixing rod according to another embodiment.
(14) FIG. **9**B is an enlarged side view illustrating a portion of an aerator mixing rod according to another embodiment.
(15) FIG. **9**C is an enlarged side view illustrating a portion of an aerator mixing rod according to another embodiment.
(16) FIG. **10** is a perspective view of a food product dispenser incorporating an aerator mixing rod.
(17) FIG. **11** is a perspective view of the food product dispenser of FIG. **10**, illustrating a dispensing unit separated from a drive unit of the food product dispenser.
(18) FIG. **12** is a cross-sectional view of the dispensing unit of FIG. **11**, illustrating a product flow path through the dispensing unit.
(19) FIG. **13** is a cross-sectional view illustrating a portion of a dispensing unit according to another embodiment and incorporating an aerator mixing rod.
DETAILED DESCRIPTION
(20) Before any embodiments of the disclosure are explained in detail, it is to be understood that the disclosure is not limited in its application to the details of construction and the arrangement of

components set forth in the following description or illustrated in the following drawings. The disclosure is capable of other embodiments and of being practiced or of being carried out in various ways.

(21) FIGS. **1-4** illustrate an aerator mixing rod **10** according to one embodiment. The aerator mixing rod **10**, (which may also be referred to as an “aerator rod,” “mixing rod,” or “static mixing rod”), includes an elongated body **12** having a first end **13** and a second end **14** opposite the first end **13** (FIG. **1**). The body **12** defines a longitudinal axis **15**, which extends centrally through the first and second ends **13**, **14**. The body **12** has a generally cylindrical shape in the illustrated embodiment; however, the body **12** may be shaped differently in other embodiments.

(22) With reference to FIGS. **3**A-**3**B, the illustrated aerator mixing rod **10** includes a plurality of teeth **20** projecting radially from an outer surface **21** of the mixing rod **10**. In the illustrated embodiment, the teeth **20** are arranged in a plurality of first rows **22**A and a plurality of second rows **22**B. The first rows **22**A and second rows **22**B alternate along the longitudinal axis **15** of the aerator mixing rod **10**.

(23) Each first row **22**A of teeth **20** defines a first plurality of passageways **30** between adjacent teeth **20** of the first row **22**A, and each second row **22**B of teeth **20** defines a second plurality of passageways **32** between adjacent teeth **20** of the second row **22**B. Each first row **22**A of teeth **20** and second row **22**B of teeth **20** are axially spaced by an interstitial distance G (FIG. **3**A) such that an axial gap is defined between the rows **22**A, **22**B of teeth **20**.

(24) With continued reference to FIGS. **3**A-**3**B, the first passageways **30** are at least partially axially misaligned with the second passageways **32**, (that is, the first passageways **30** are offset from the second passageways **32** in a circumferential direction of the aerator mixing rod **10**), to define a serpentine or tortuous flow path (see e.g., FIG. **3**B) through the passageways **30**, **32** along the exterior of the aerator mixing rod **10**. In other words, there is no direct axial flow path that would allow a liquid product to flow directly through the first passageways **30** and the second passageways **32** without impinging upon the teeth **20**.

(25) Referring to FIGS. **3**A and **3**B, the first and second rows **22**A, **22**B of teeth **20** can be configured to have four, six, or eight teeth in each respective row **22**A, **22**B. For example, each first row **22**A of teeth **20** may have four teeth **20**, and each second row **22**B of teeth **20** may have eight teeth **20**. In the illustrated embodiment, the first and second rows **22**A, **22**B each have the same number of teeth **20**. For example, the first and second rows **22**A, **22**B may each include eight teeth (FIG. **4**).

(26) In some embodiments, the aerator mixing rod **10** may include twenty or more rows of teeth **20** (e.g., ten first rows **22**A and ten second rows **22**B). In other embodiments, the aerator mixing rod **10** may include thirty or more rows of teeth **20** (e.g., fifteen first rows **22**A and fifteen second rows **22**B). In other embodiments, the aerator mixing rod **10** may include forty or more rows of teeth (e.g., twenty first rows **22**A and twenty second rows **22**B). In yet other embodiments, the aerator mixing rod **10** may include a greater number of unique rows (e.g., three rows, four rows, or more) arranged in various repeating patterns to provide a tortuous flow path with desired characteristics to suit a particular product or application.

(27) In operation, the aerator mixing rod **10** is positioned within a fluid flow passage, such as a tube. A liquid product is moved through the fluid flow passage (e.g., by a pump, pressurized gas source, or any other means). When the liquid product encounters the aerator mixing rod **10**, the liquid product is forced through the passageways **30**, **32** between the teeth **20**. As the liquid product travels the length of the aerator mixing rod **10**, the liquid product is sheared by the teeth **20**, effectively mixing and, in some embodiments, aerating and whipping the liquid product and producing a foam-like airy substance. The serpentine flow pattern (FIG. **3**B) facilitates the creation of a proper whipped product yielding an overall desirable appearance and consistency.

(28) For food product whipping applications, the aerator mixing rod **10** is formed in a mold assembly from a food-safe, non-porous, injection moldable material. In the illustrated embodiment,

the aerator mixing rod **10** is formed from an injection moldable thermoplastic material, and more particularly an engineering thermoplastic such as Delrin® acetal homopolymer, to provide the aerator mixing rod **10** with sufficient strength and dimensional stability. The aerator mixing rod **10** may alternatively be made from other injection moldable materials, including but not limited to aluminum or stainless steel (e.g., via metal injection molding), ceramics, other plastics (e.g. PBT, PET, PTT, PLA, PP, ABS, ASA, PEI), cellulose-based materials or other biomaterials, silicone-based compounds, or foamed materials (e.g., polyurethane, expanded polystyrene).

(29) In some embodiments, the aerator mixing rod **10** may be formed from a combination of materials via multiple molding and/or coating operations. For example, with reference to FIGS. **2** and **4**, the illustrated aerator mixing rod **10** includes a core **40** and an outer shell **42** surrounding the core **40**. The core **40** may be formed via a first molding operation, and the outer shell **42** may subsequently be molded around the core **40** in a second molding operation. In the illustrated embodiment, the teeth **20** are formed as part of the outer shell **42**.

(30) Forming the aerator mixing rod **10** using a multi-step molding process provides advantages, including a reduced mold volume per molding process, which in turn improves dimensional stability by reducing contraction that occurs during solidification and cooling. A multi-step molding process also allows the aerator mixing rod **10** to optionally be formed from multiple different materials. For example, in some embodiments, the core **40** may be formed from a foamed material, and the outer shell **42** may be formed from a thermoplastic material. Such embodiments may allow the aerator mixing rod **10** to be lighter weight and have other properties that may be desirable in certain applications, such as low thermal conductivity.

(31) FIGS. **5**A-**5**B illustrate exemplary mold assemblies **200**, **250** that may be used to form the aerator mixing rod **10**. Each of the illustrated mold assemblies **200**, **250** includes a plurality of movable mold parts with different release configurations. For example, the illustrated mold assembly **200** has a symmetrical release configuration with four segmented mold parts **210** configured to abut one another at respective parting lines **215***a*, **215***b* (FIG. **5**A). In the illustrated embodiment, the parting lines **215***a*, **215***b* are transverse and intersect at a center of the mold cavity formed by the mold parts **210** (corresponding to the center of the aerator mixing rod **10**). The symmetrical configuration of the mold assembly **200** allows the mold parts **210** to all be released from their respective parting lines **215** in radial draw directions **220**.

(32) Thus, a method of manufacturing the aerator mixing rod **10** using injection molding may include arranging the mold parts **210** into a mold assembly structure **200** that defines a mold cavity shaped to form the aerator mixing rod **10**. Next, the method may further include heating an injection moldable material, injecting the injection moldable material into the mold assembly structure **200** to form the aerator mixing rod **10**, and, after a cooling period, removing the aerator mixing rod **10** from the mold assembly structure **200** by separating the plurality of mold parts **210** from the aerator mixing rod **10** along the parting lines **215**. The mold parts **210** move in the radial draw directions **220** during separation.

(33) With reference to FIG. **5**B, the illustrated mold assembly **250** has an asymmetrical release configuration that includes two first mold parts **260** defining respective parting lines **265***a*, **265***b* and two second mold parts **270** defining respective parting lines **285***a*, **285***b*. With reference to the orientation illustrated in FIG. **5**B, the mold assembly **250** allows the first mold parts **260** to be released vertically (in the direction of arrows **280**) from the mold parting lines **265***a*, **265***b*. Either simultaneously or in sequence, the second mold parts **270** can be released horizontally (in the direction of arrows **275**) from the parting lines **285***a*, **285***b*.

(34) Thus, a method of manufacturing the aerator mixing rod **10** using injection molding may include arranging the mold parts **260**, **270** into a mold assembly structure **250** that defines a mold cavity shaped to form the aerator mixing rod **10**. Next, the method may further include heating an injection moldable material, injecting the injection moldable material into the mold assembly structure **250** to form the aerator mixing rod **10**, and, after a cooling period, removing the aerator

mixing rod **10** from the mold assembly structure **250** by separating the mold parts **260** along the parting lines **265***a*, **265***b* in the draw directions **280**, and separating the mold parts **270** along the parting lines **285***a*, **285***b* in the draw directions **275**.

(35) As illustrated in FIGS. **5**A and **5**B, the symmetrical mold assembly **200** includes two part lines **215***a*, **215***b*, whereas the asymmetrical mold assembly **250** includes four part lines **265***a*, **265***b*, **285***a*, **285***b*. In other embodiments, other mold release configurations may be used. That is, the aerator mixing rod **10** may be formed using mold assemblies having other numbers and/or configurations of mold parts to facilitate different constructions of the aerator mixing rod **10**. As described in greater detail below, the mold assembly used to form the aerator mixing rod is configured to prevent undercuts that would inhibit releasing the mold parts (e.g., **210**, **260**, **270**) from the aerator mixing rod **10** after molding. Undercuts are any surfaces that cannot be seen when viewed from the pull direction of the mold part. If an undercut feature is molded, the mold part may catch on the undercut, which prevents the mold part from retracting after molding.

(36) For example, FIG. **6**A illustrates an example of an aerator mixing rod **10**A, which cannot be molded using the asymmetrical mold assembly **250** of FIG. **5**B without forming undercuts. More specifically, due to the geometry of the aerator mixing rod **10**A illustrated in FIG. **6**A, multiple undercuts **300** exist that would prevent the first mold parts (not shown) from retracting in the direction of the arrows **280**. To avoid the undercuts **300**, the same aerator mixing rod **10**A can be molded using the symmetrical mold assembly **200** of FIG. **5**A. That is, the aerator mixing rod **10**A can be molded without any undercuts **300** by radially releasing the mold parts (not shown) in the direction of arrows **220**.

(37) The minimum number of mold parts to form an aerator mixing rod, such as the aerator mixing rod **10**, may be proportional to the greatest number of teeth **20** in a respective row **22**A, **22**B. For example, in embodiments in which the rows **22**A, **22**B each have a maximum of four teeth **20**, two mold parts arranged to separate from a single part line may be used to form the desired structure of the aerator mixing rod **10**. In embodiments in which the rows **22**A, **22**B each have a maximum of six teeth **20**, three mold parts may be used to form the desired structure of the aerator mixing rod **10**, and in embodiments in which the rows **22**A, **22**B each have a maximum of eight teeth **20**, four mold parts may be used to form the desired structure of the aerator mixing rod **10**. As such, in some embodiments, the number of mold parts of the mold assembly may be equal to two times the maximum number of teeth **20** in any given row along the aerator mixing rod **10**. Although each of the examples described herein includes an even number of teeth **20** in the respective rows **22**A, **22**B, in other embodiments, the aerator mixing rod **10** may be configured with an odd number of teeth **20** in one or both rows **22**A, **22**B.

(38) The aerator mixing rod **10** can be formed with teeth **20** of various geometries. For example, FIGS. **7**A and **7**B illustrate aerator mixing rods **10**B and **10**C having different tooth geometries. An outer profile **310** of the teeth **20** defines a first diameter D**1** (i.e. tip-to-tip), and an inner profile or base **320** of the teeth **20** defines a second diameter (i.e. base-to-base) D**2**. Thus, the height of the teeth **20** may be expressed as a ratio of the first and second diameters D**1**:D**2**.

(39) With continued reference to FIGS. **7**A and **7**B, the geometry of each tooth **20** may be further defined by a series of angles all measured with respect to a radial reference line or axis **300** that extends through the center of the tooth **20**. FIGS. **7**A and **7**B illustrate a first angle, α (alpha), defined between the axis **300** and an inner side edge of an adjacent tooth **20**, a second angle, γ (gamma), defined between the axis **300** and an inner side edge of the tooth **20**, and a third angle, β (beta), defined between the axis **300** and an outer side edge of the tooth **20**. Finally, referring to FIG. **3**A, each tooth **20** defines a thickness T of in an axial direction (i.e. parallel to the axis **15** of the aerator mixing rod **10**).

(40) The thickness T and interstitial distance G (FIG. **3**A), and the angles α, γ, β diameters D**1**, D**2** (FIGS. **7**A-**7**B), may be optimized for a desired mixing application or material. For example, varying the angle α varies the offset-distance between adjacent teeth **20** and thus the areas of the

respective passageways **30**, **32** between adjacent teeth **20**. Additionally, varying the diameters D**1**, D**2**, and/or the angles α, γ, β allows an overlap between the rows of teeth **20**, **22** to be optimized for a specific mixing material or application. For example, by increasing β, the width of each tooth at the outer profile **310** will increase, resulting in an increased amount of tooth overlap between the plurality of teeth **20** on the aerator mixing rod **10**. A greater tooth overlap may provide a more tortuous path between the rows **22**A, **22**B of teeth **20**, producing a greater mixing and/or aerating effect. The angle γ may be increased to make the teeth **20** more v-shaped (e.g., FIG. **7**B) or decreased to make the teeth **20** more rectangular (e.g., FIG. **7**A).

(41) In addition to determining tooth overlap to ensure proper mixing, the reference angles α, γ, β control the axial surface area of each tooth **20**. The shear force acting on liquid product that flows along the aerator mixing rod **10** is dependent upon the surface areas of the teeth **20**. A greater surface area of the teeth **20** may also increase the flow-resistance and the corresponding pressure drop across the aerator mixing rod **10**.

(42) Thus, the geometry of an aerator mixing rod is complex, and the dimensions and angles described above each play a role in the performance of an aerator mixing rod in a particular application. Through extensive testing and analysis, the following ranges and ratios have been discovered to provide optimum performance when using the aerator mixing rod **10** to produce a whipped cream product (including both dairy and non-dairy based cream).

(43) More specifically, in some embodiments, the ratio D**1**:D**2** may be between 10:1 and 20:19, such that D**2** is between 10% of D**1** and 95% of D**1**. In other embodiments, the ratio D**1**:D**2** may be between 5:1 and 20:19, such that D**2** is between 20% of D**1** and 95% of D**1**. In yet other embodiments, the ratio D**1**:D**2** may be between 3:1 and 10:9, such that D**2** is between 33% of D**1** and 90% of D**1**. In embodiments in which the aerator mixing rod **10** is used for whipping cream, it has been found that a ratio of D**1**:D**2** between 14:10 and 14:13, such that D**2** is between about 71% and about 93% of D**1** may be particularly advantageous.

(44) In some embodiments, the thickness T of each tooth **20** may be equal to the interstitial distance G between the rows **22**A, **22**B of teeth **20**. In other embodiments, the ratio of the thickness T of each tooth **20** the interstitial distance G can be between 0.5:1 and 1:2. In other words, the interstitial distance G can be between half the thickness T to twice the thickness T of each tooth **20**. It has been found that this range of ratios of the thickness T to the interstitial distance G may be particularly advantageous for whipping cream.

(45) In some embodiments, the angle γ is greater than or equal to 12°. For example, the angle γ can be 12°, 16°, 20°, 22.5, or other angles. The angle β can be any angle between 0° and 20° in some embodiments. For example, the angle β can be 5°, 11.25°, 15°, or other angles. In some embodiments, the angle β can be any angle between 0° and 11.25°. It has been found that these values and ranges for the angles γ and β may be particularly advantageous for whipping cream.

(46) Importantly, in addition to affecting the aeration/texture of the final whipped product, the angles α, γ, β also affect whether the aerator mixing rod **10** is injection moldable. First, referring to FIG. **7**B, the angle γ must be greater than the angle β such that the inner width of the tooth **20** is wider than the outer width. In other words, the teeth **20** are tapered in a radial direction. This allows undercuts to be avoided. In some embodiments, the angle γ is at least 0.5° greater than the angle β. In some embodiments, the angle γ is at least 2° greater than the angle β. In some embodiments, the angle γ is at least 5% greater than the angle β. In some embodiments, the angle γ is at least 10% greater than the angle β.

(47) In some embodiments, the teeth **20** of the aerator mixing rod **10** may include draft angles to make the aerator mixing rod **10** more easily removable from the mold assembly. For example, FIG. **9**B illustrates an aerator mixing rod **10**F in which the axial faces of each tooth **20** define draft angles θ relative to an axis transverse to the longitudinal axis **15** of the aerator mixing rod **10**F. In the illustrated embodiment, the draft angles θ are each greater than 0° (e.g. 1° or more) to facilitate the release of the mixing rod **10** from the mold assembly. In some embodiments, the draft angles θ

on each axial side **20***s* of the teeth **20** may be the same or different.

(48) Typically, parts manufactured using CNC machining have a draft angle θ of zero degrees, because CNC machining utilizes a cutting tool to shape the specific features of the part, rather than a molten plastic (or other injection moldable material) being injected into a mold, which must then be released and removed from the mold. When a part is being removed from a mold, a friction force exists between the mold and the part that can tear or deform the part. With the addition of a draft angle θ, the friction force is reduced due to the tapered "v" shape of the part. In some embodiments, the draft angle θ can be proportional to the tooth height H (FIG. **9**B). For example, the larger the tooth height H of the plurality of teeth **20**, the larger the draft angle θ.

(49) With reference to FIG. **8**, in some embodiments, the aerator mixing rod **10** can be formed from a plurality of longitudinal segments. For example, FIG. **8** illustrates an aerator mixing rod **10**D split into a first portion **60** and a second portion **70** configured to be joined together to form the full length of the aerator mixing rod **10**D. Because each portion **60**, **70** is only half of the length of the aerator mixing rod **10**D, separately molding the first portion **60** and the second portion **70** may produce less tolerance stack-up and therefore result in greater dimensional accuracy and/or lower precision requirements for the mold assembly. In some embodiments, the first portion **60** and the second portion **70** may include cooperating attachment features **65**, **68** (e.g., a projection **65** on the first portion **60** and a corresponding recess **68** on the second portion **70**, or vise versa) to facilitate coupling the first portion **60** to the second portion **70**. In other embodiments, the first portion **60** and the second portion **70** can be separately injection moldable parts arranged side-by-side, rather than being joined in an axial direction.

(50) In some embodiments, the aerator mixing rod **10** may include relief gaps between adjacent teeth. For example, FIG. **9**A illustrates an aerator mixing rod **10**E including a relief gap **400** between adjacent teeth **20** in the circumferential direction. In other words, the teeth **20** in each row **22**A, **22**B are not continuously connected. Rather, a portion of the outer surface **21** of the main body **12** of the aerator mixing rod **10**E extends between adjacent teeth **20**. In other embodiments, a diameter of the relief gap **400** measured from the center of the aerator mixing rod **10** and the diameter of the outer surface **21** of the aerator mixing rod **10** can form a ratio between the ranges of 1:1-12:9. That is, in some embodiments, adjacent teeth in the circumferential direction may be interconnected by a raised wall (not shown) extending across the relief gap **400**.

(51) In some embodiment, the teeth **20** of the aerator mixing rod **10** may have a varying height. For example, FIG. **9**C illustrates an aerator mixing rod **10**G with teeth **20** configured such that an outer surface **800** of each tooth **20** is angled with respect to the longitudinal axis **15** of the aerator mixing rod **10**G. The teeth **20** thus form a raised edge **810**, which defines the maximum height H of the tooth. The angled surfaces **800** may produce greater turbulence in the liquid product as the product flows past the teeth **20**, which may advantageously enhance mixing for certain products. In some embodiments, the angled surface **800** may be oriented at an angle between 1 degree and 45 degrees relative to the longitudinal axis **15**. In other embodiments, the angled surface **800** may be oriented at an angle between 5 degrees and 30 degrees relative to the longitudinal axis **15**.

(52) FIG. **10** illustrates a food product dispenser **100**. The food product dispenser **100** may be configured to utilize an injection moldable aerator mixing rod, such as the aerator mixing rod **10**. The illustrated food product dispenser **100** includes a drive unit **114** and a dispensing unit or module **118** removably coupled to the drive unit **114**. The dispensing unit **118** includes a product reservoir **120** containing the liquid product to be whipped, a dispensing nozzle **122**, and a product transfer assembly or whipping assembly **126** configured to be powered by the drive unit **114** to move product from the reservoir **120** to the dispensing nozzle **122**.

(53) Referring to FIG. **11**, the illustrated drive unit **114** includes a drive shaft **130** configured to engage a drive socket **132** on the whipping assembly **126** when the dispensing unit **118** is coupled to the drive unit **114**. The drive shaft **130** is driven by a motor (not shown) housed within the drive unit **114** to provide a rotational input to the whipping assembly **126**.

(54) The whipping assembly **126** includes an aerator in fluid communication with the product reservoir **120** and a pump assembly **146** (e.g., a gear pump, wiper pump, or the like) driven by the motor (via the drive shaft **130** and drive socket **132**) for drawing the product from the product reservoir **120** and forcing the product through the aerator to form an aerated or “whipped” product. The aerator communicates with the dispensing nozzle **122**, which is configured to dispense the whipped product.
(55) In some embodiments, the dispensing unit **118** may include the motor. In such embodiments, the drive shaft **130** and drive socket **132** may be replaced by electrical connectors. In other embodiments, the drive unit **114** may include a source of pressurized gas, such as a refillable and/or interchangeable pressurized gas canister, and/or a compressor operable to generate pressurized gas on demand. In such embodiments, the drive shaft **130** and drive socket **132** may be replaced by a pneumatic connector, and preferably a quick-release pneumatic connector such as a bayonet fitting. The drive unit **114** may then supply the pressurized gas to the dispensing unit **118** to force the liquid product from the product reservoir **120** through the aerator (e.g., by pressurizing the product reservoir **120**). Alternatively, the pump may include a rotary vane, and the pressurized gas may drive the rotary vane to operate the pump. In yet other embodiments, the pressurized gas may be directed through a venturi, creating suction to draw liquid product from the product reservoir. The liquid product may then be entrained in the flow of pressurized gas and directed through the aerator.
(56) With reference to FIG. **11**, the dispensing unit **118**—which includes the product reservoir **120**, whipping assembly **126**, and dispensing nozzle **122**—can be quickly removed from the drive unit **114** as a single, self-contained assembly. This allows a user to remove the dispensing unit **118** when not in use and store it in a refrigerator. The product and all of the downstream components that contact the product can therefore be maintained at safe temperatures without requiring a dedicated refrigeration system. This advantageously reduces the size, cost, complexity, energy requirements, and operating noise of the dispenser **100** as compared to existing dispensers with on-board refrigeration systems.
(57) The product reservoir **120** of the dispensing unit **118** is preferably insulated in order to keep the product contained therein at a suitably cold temperature for a long period of time when the dispensing unit is outside of the refrigerator. For example, the product reservoir **120** may be a double-walled, vacuum-insulated canister. The product reservoir **120** may be made of stainless steel, or any other insulating, food-safe material, including but not limited to a plastic material. In some embodiments, the product reservoir **120** may include a thermally-conductive area in contact with an inner wall of the product reservoir **120** to enhance cooling of the product within the reservoir **120** when the dispensing unit **118** is placed in the refrigerator. In such embodiments, an insulating cover may be provided to cover the thermally-conductive area when the product reservoir **120** is removed from the refrigerator for use. In some embodiments, the thermally conductive area may be cooled by ice or a cooling apparatus (such as a thermoelectric cooler) while the dispensing unit **118** is coupled to the drive unit **114**.
(58) In some embodiments, the product reservoir **120** may be a disposable product package, such as an aseptic brick package, a plastic or metal foil pouch, or a bag-in-box assembly. Disposable product packaging may facilitate interchanging the type of product to be dispensed by the dispensing unit **118** without having to clean the product reservoir **120**. In any such embodiments, the product reservoir **120** may optionally be insertable into an insulating sleeve or casing.
(59) With reference to FIG. **12**, the whipping assembly **126** includes a housing **152** removably coupled to the product reservoir **20**, with an aerator housing portion **170** extending into the product reservoir **120**. The aerator housing portion **170** includes a first chamber **172** and a second chamber **174** separated by a longitudinally-extending dividing wall **175**. The second chamber **174** is in fluid communication with the first chamber **172** via a transfer passage **176** extending through the dividing wall **175**.

(60) In the illustrated embodiment, the transfer passage **176** includes a first rounded bore **176***a* and a second rounded bore **176***b* intersecting the first rounded bore **176***a*. The rounded bores **176***a*, **176***b* may have generally spherical profiles. In some embodiments, the first rounded bore **176***a* is formed by inserting a ball end mill through a bottom end of the aerator housing portion **170** and into the first chamber **172** until the ball end mill engages and removes material from the dividing wall **175**. Likewise, the second rounded bore **176***b* is formed by inserting the ball end mill through the bottom end of the aerator housing and into the second chamber **174** until the ball end mill engages and removes material from the dividing wall **175** opposite the first rounded bore **176***a*. Machining the transfer passage **176** in this manner advantageously allows for the transfer passage **176** to be formed without requiring any additional access openings, which would be required to drill transversely through the dividing wall **175** using a straight drill bit, for example. In addition, the rounded bores **176***a*, **176***b* lack sharp corners and 90-degree interface angles, which inhibits product from becoming lodged in the transfer passage **176** and thereby facilitates cleaning. In some embodiments, the transfer passage **176** (including the rounded bores **176***a*, **176***b*) may be formed in other ways, including but not limited to injection-molding or 3D printing.
(61) With continued reference to FIG. **12**, a first mixing rod **178** is supported within the first chamber **172**, and a second mixing rod **180** is supported within the second chamber **174**. In the illustrated embodiment, the first and second mixing rods **178**, **180** are stationary labyrinth mixing rods, such as the aerator mixing rod **10** described above. As such, the mixing rods **178**, **180**, have teeth **20** and passageways **30**, **32** (FIGS. **3**A-**3**B) to define a tortuous flow pathway along the exterior of the mixing rods **178**, **180**. In other embodiments, one or more mixing rods of other types or geometries may be used. In the illustrated embodiment, each of the mixing rods **178**, **180** is made of plastic; however, the mixing rods **178**, **180** may be made from other materials in other embodiments.
(62) In use, the drive unit **114** drives the pump assembly **146**, which forces an air and product mixture into the first chamber **172** of the aerator housing portion **170**. The air and product mixture then flows along the first mixing rod **178** in a first direction (i.e. the direction of arrow A as shown in FIG. **12**), which partially aerates the product. Upon reaching the end of the first mixing rod **178**, the partially aerated product flows through the transfer passage **176** in a second direction. In the illustrated embodiment, the second direction is generally transverse to the first direction. The partially aerated product then flows in a third direction (i.e. in the direction of arrow B), which is generally opposite the first direction, and over the second mixing rod **180**. This completes aeration of the product, and the aerated or whipped product is discharged from the second chamber **174** through the dispensing nozzle **122**.
(63) By providing two mixing rods **178**, **180** in separate sections, the overall height of the assembly is reduced, which in turn allows the overall size of the dispensing unit **118** to be minimized. In addition, the manufacturing tolerances for the mixing rods **178**, **180** may be reduced, since the relatively shorter length of each rod **178**, **180** (compared to a single-piece rod having a length equal to the combined lengths of the rods **178**, **180**) produces less tolerance stack-up. In other embodiments, however, other mixing rod configurations may be used, including a single-piece mixing rod, or any other number of mixing rods.
(64) During operation, shearing of the product mixture that takes place as the product mixture flows over the mixing rods **178**, **180** produces heat. Because the mixing rods **178**, **180** are made of a material with low thermal conductivity (e.g., plastic in the illustrated embodiment), a minimal amount of heat is absorbed by the mixing rods **178**, **180**. Rather, the generated heat is carried away with the product. In the illustrated embodiment, the mixing rods **178**, **180** have a thermal conductivity between 0.1 and 0.5 Watts/Meter-Kelvin. In contrast, a conventional mixing rod, which is typically made of metal such as stainless steel, may have a thermal conductivity between 10 and 20 Watts/Meter-Kelvin or more. Thus, a conventional mixing rod may have a thermal conductivity at least 50 to 100 times greater than the mixing rods **178**, **180**, resulting in more heat

being absorbed by the mixing rod. The low thermal conductivity of the mixing rods **178**, **180** in the illustrated embodiment is particularly advantageous when the housing portion **170** is submerged within product contained within the product reservoir **120**, such that heating of the product within the product reservoir **120** is minimized.

(65) FIG. **13** illustrates a dispenser **100**″ according to another embodiment. The dispenser **100**″ is similar to the dispenser **100** described above with reference to FIGS. **10-12** but includes an aerator mixing rod **10**″ received within a housing **170**″ and having a nested configuration. In particular, the aerator mixing rod **10**″ includes a first portion **60**″ and a second portion **70**″ surrounding and arranged concentrically with the first portion **60**″. The second portion **70**″ includes a plurality of teeth (such as the teeth **20**) located on both an interior side **71**″ and an exterior side **72**″ of the second portion **70**″. In some embodiments, the teeth on the exterior side **72**″ of the second portion **70**″ may alternatively be located on an inner wall of the housing **170**″. In some embodiments, the teeth on the interior side **71**″ of the second portion **70**″ may alternatively be located on an exterior side **62**″ of the first portion **60**″.

(66) In operation, liquid product is forced through a tortuous pathway defined between the first portion **60**″ and the second portion **70**″. Upon reaching a bottom end **171**″ of the housing **170**″, the partially-whipped product is redirected outside of the second portion **70**″. The product then flows in the opposite direction, through the tortuous pathway between the exterior side **72**″ of the second portion **70**″ and the inner wall of the housing **170**″. This completes aeration of the product, and the aerated or whipped product is then discharged through a dispensing nozzle **122**″.

(67) The aerator mixing rods described and illustrated herein can be used to produce different product mixtures, such as air and water, oil and air, solutes and solvents, etc. It should be understood that, in addition to food product dispensers, an injection-molded aerator mixing rod may advantageously be used in a wide variety of applications. Finally, although the aerator mixing rods described and illustrated herein are preferably made by injection molding, in other embodiments, the aerator mixing rods may be manufactured by other methods, including but not limited to 3D printing or other additive manufacturing processes. Manufacturing aerator mixing rods using such alternative methods may suitable for small volume productions.

(68) Although the disclosure has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the scope and spirit of one or more independent aspects of the disclosure as described.

(69) Various features and aspects of the disclosure are set forth in the following claims.

Claims

1. A method of manufacturing an aerator mixing rod using injection molding, the method comprising: arranging a plurality of mold parts into a mold assembly structure, the mold assembly structure defining a mold cavity shaped to form the aerator mixing rod, such that the aerator mixing rod includes a body having a first end, a second end opposite the first end, and an outer surface extending between the first and second ends, the body defining a longitudinal axis extending through the first and second ends, and a plurality of teeth extending radially outward from the outer surface of the body. the plurality of teeth including a first row of teeth and a second row of teeth spaced from the first row of teeth along the longitudinal axis, wherein a first passageway is formed between adjacent teeth of the first row of teeth, and a second passageway is formed between adjacent teeth of the second row of teeth, wherein the first passageway is at least partially misaligned with the second passageway in a direction parallel to the longitudinal axis such that the first passageway and the second passageway are configured to at least partially form a tortuous path for fluid flowing along the outer surface of the body; heating an injection moldable material; injecting the injection moldable material into the mold assembly structure to form the aerator mixing rod; and after a cooling period, removing the aerator mixing rod from the mold assembly

structure by separating the plurality of mold parts from the aerator mixing rod, wherein arranging the plurality of mold parts into the mold assembly structure includes selecting between a plurality of different mold release configurations in order to prevent an undercut, wherein at least one of the mold release configurations is a symmetrical configuration including a plurality of part lines intersecting at a center of the mold cavity, and wherein separating the plurality of mold parts includes separating the mold parts along the part lines.

2. The method of claim 1, wherein each tooth of the plurality of teeth includes a first side and a second side generally opposite the first side, and wherein the first and second sides converge in a radial outward direction of the aerator mixing rod.

3. The method of claim 2, wherein the first and second sides are axially-facing sides.

4. The method of claim 2, wherein the first and second sides are circumferentially facing sides.

5. The method of claim 2, wherein each tooth of the plurality of teeth includes an inner width and an outer width, the inner width defined adjacent the outer surface of the body, and wherein the inner width is greater than the outer width.

6. The method of claim 5, wherein the inner width is at least 5% greater than the outer width.

7. The method of claim 2, wherein each tooth of the plurality of teeth includes an outermost point, wherein the outermost points of the plurality of teeth collectively define a first diameter, wherein the body defines a second diameter, and wherein the second diameter is between 10% and 95% of the first diameter.

8. The method of claim 2, wherein the injection-moldable material is a plastic material.

9. The method of claim 8, wherein the plastic material includes an acetal homopolymer.

10. The method of claim 2, wherein the first row of teeth and the second row of teeth each include 8 teeth.

11. The method of claim 2, wherein the aerator mixing rod includes a core and an outer shell, wherein the outer shell is molded over the core, and wherein the outer shell includes the plurality of teeth.

12. The method of claim 1, wherein the aerator mixing rod includes a thermal conductivity between 0.1 and 0.5 Watts/Meter-Kelvin.

13. A method of manufacturing an aerator mixing rod using injection molding, the method comprising: arranging a plurality of mold parts into a mold assembly structure, the mold assembly structure defining a mold cavity shaped to form the aerator mixing rod, such that the aerator mixing rod includes a body having a first end, a second end opposite the first end, and an outer surface extending between the first and second ends, the body defining a longitudinal axis extending through the first and second ends, and a plurality of teeth extending radially outward from the outer surface of the body, the plurality of teeth including a first row of teeth and a second row of teeth spaced from the first row of teeth along the longitudinal axis, wherein a first passageway is formed between adjacent teeth of the first row of teeth, and a second passageway is formed between adjacent teeth of the second row of teeth, wherein the first passageway is at least partially misaligned with the second passageway in a direction parallel to the longitudinal axis such that the first passageway and the second passageway are configured to at least partially form a tortuous path for fluid flowing along the outer surface of the body; heating an injection moldable material; injecting the injection moldable material into the mold assembly structure to form the aerator mixing rod; and after a cooling period, removing the aerator mixing rod from the mold assembly structure by separating the plurality of mold parts from the aerator mixing rod, wherein arranging the plurality of mold parts into the mold assembly structure includes selecting between a plurality of different mold release configurations in order to prevent an undercut, wherein at least one of the mold release configurations is an asymmetrical configuration including four part lines, and wherein separating the plurality of mold parts includes separating the mold parts along the part lines.

14. The method of claim 13, wherein the injection-moldable material is a plastic material.

15. The method of claim 14, wherein the injection-moldable material includes an acetal homopolymer.